Node.js Path Module - Full Roman Urdu Revision

Personal detailed notes - tumhari aaj ki practice ke mutabiq

Aaj tumne Node.js ka built-in `path` module practice kiya. Iska basic kaam files aur folders ke paths ko safely handle karna hai. Tumne do tasks mein join, resolve, basename, extname, dirname, parse aur sep use kiye; extra practice mein isAbsolute bhi use kiya.

1. Path module ko load kaise karna hai?

```
const path = require("path");
```

Yahan `require("path")` Node.js ke built-in Path module ko hamari file mein lata hai. Isko npm se install karne ki zarurat nahi hoti.

2. path.join() - path ke pieces ko jorna

Agar tumhare paas alag-alag folder/file names hon, `join()` unko mila kar proper path bana deta hai.

```
const filePath = path.join("projects", "nodejs", "backend", "server.js");
```

Simple yaad rakhna: **JOIN = pieces ko joro.** Tumne Task 1 mein students/junaid/nodejs/practice.js aur Task 2 mein projects/nodejs/backend/server.js banaya tha.

3. path.resolve() - absolute path

`resolve()` path ko resolve karke absolute/full path deta hai. Agar tum relative path do, to current working location ko use karke full path ban sakta hai.

```
const absolutePath = path.resolve(filePath);
console.log(absolutePath);
```

Shortcut: join = path banana/join karna; resolve = us path ko absolute form tak resolve karna.

4. path.basename() - akhir mein kya hai?

Ye path ka last part deta hai. File path ho to aam tor par file ka naam milta hai.

```
path.basename(filePath); // server.js
```

BASE NAME = file ka poora naam including extension.

5. path.extname() - extension nikalna

File kis type ki lag rahi hai uski extension nikalne ke liye `extname()` use hota hai.

```
path.extname(filePath); // .js
```

Examples: server.js -> .js, notes.pdf -> .pdf, image.png -> .png.

6. path.dirname() - file kis directory mein hai?

`dirname()` final file name ko side par rakh kar us se pehle wali directory path return karta hai.

```
path.dirname(filePath);
```

Yaad rakho: **DIR NAME = directory wala hissa.**

7. path.parse() - ek path ki complete breakdown

Ye tumhare liye important concept hai. `parse()` ek path leta hai aur uski information ko **object** mein return karta hai.

```
const parsedPath = path.parse(filePath);
```

Is returned object mein commonly ye properties hoti hain:

root - root part
dir - directory
base - file name + extension
ext - sirf extension
name - extension ke baghair file name

```
console.log(parsedPath.name); // server
console.log(parsedPath.base); // server.js
console.log(parsedPath.ext);  // .js
console.log(parsedPath.dir);
```

Tum Task 2 mein pehle `ext` aur `dir` nikal chuke thay; phir missing `name` aur `base` bhi khud add kiye. Is liye parse object property access ka concept bhi practice ho gaya.

8. name vs base - confuse mat hona

Property	server.js par result
name	server
base	server.js
ext	.js

9. path.sep - separator kya hota hai?

`path.sep` current operating system ka path separator deta hai.

```
console.log(path.sep);
```

Windows mein tum aksar backslash dekhoge aur Unix-like systems mein forward slash. Is liye Node ka path module OS ke path rules handle karne mein useful hai.

10. path.isAbsolute() - tumhari extra practice

Ye check karta hai ke diya gaya path already absolute hai ya nahi. Result Boolean hota hai: true ya false.

```
const check = path.isAbsolute(filePath);
console.log(check);
```

Isko tumne Task 2 mein apni taraf se extra add kiya tha. Important difference: `resolve()` path resolve karta hai, jabke `isAbsolute()` sirf check karta hai.

11. Ek line mein sab yaad karo

join = joro
resolve = absolute/full resolve karo
basename = last/file name
extname = extension
dirname = directory
parse = complete breakdown object
sep = OS separator
isAbsolute = absolute hai ya nahi check karo

12. Tumhari practice jaisa complete example

```
const path = require("path");

const filePath = path.join("projects", "nodejs", "backend", "server.js");

console.log("Full Path:", filePath);
console.log("Absolute:", path.resolve(filePath));
console.log("Base Name:", path.basename(filePath));
console.log("Extension:", path.extname(filePath));
console.log("Directory:", path.dirname(filePath));

const parsed = path.parse(filePath);
console.log("Name:", parsed.name);
console.log("Base:", parsed.base);
console.log("Ext:", parsed.ext);
console.log("Dir:", parsed.dir);

console.log("Separator:", path.sep);
console.log("Is Absolute:", path.isAbsolute(filePath));
```

13. Common mistakes jo tumhe avoid karni hain

1. Method spelling galat na likhna: `dirname`, `basename`, `extname`.
2. File name ki spelling check karna, jaise `practice.js`.
3. `parse()` ka result object hai, is liye `parsed.name` jaisi properties access kar sakte ho.
4. `path.sep` property hai; isko `path.sep()` ki tarah function call nahi karna.
5. `resolve()` aur `isAbsolute()` same kaam nahi karte.

14. Self Test - bina notes dekhe jawab do

1. Path module require karne ki line likho.
2. 4 path pieces ko combine karne ke liye kya use karoge?
3. server.js se .js kaise niklega?
4. server.js se server kaise niklega using parse?
5. parse().base kya dega?
6. dirname ka kaam kya hai?
7. resolve aur join mein basic difference kya hai?
8. path.sep function hai ya property?
9. isAbsolute ka result kis data type ka hota hai?
10. parse() kis data type ka result return karta hai?

15. Next-day 60-second revision

Kal revision ke waqt pehle bina code dekhe bolo: **join jorta hai, resolve absolute karta hai, basename file name, extname extension, dirname directory, parse object breakdown, sep separator, isAbsolute check.** Phir ek chhota path khud bana kar har method ek dafa run kar lena. Agar ye bina help ke ho jaye to Path Module ka current lesson clear hai.